

UART Peripheral Register Map (LSO Configuration)

1. Control Registers

Offset	Name	Width	Bit Descriptions
0x00	BRR	16-bit	Baud Rate Register: 16-bit divisor for bit-timing generation.
0x02	CR1	8-bit	Control Register 1: • Bit 7: IE_RXNE (RXNE Interrupt Enable) • Bit 6: IE_TXE (TXE Interrupt Enable) • Bit 5: PS (Parity Selection: 0=Even, 1=Odd) • Bit 4: PCE (Parity Control Enable) • Bit 3: M (Word Length: 0=8b, 1=9b) • Bit 2: RE (Receiver Enable) • Bit 1: TE (Transmitter Enable) • Bit 0: UE (USART Enable)

2. Status Register

Offset	Name	Width	Bit Descriptions (Status Flags)
0x04	SR	8-bit	Status Register: • Bit 7: NE (Noise Error) • Bit 6: FE (Framing Error) • Bit 5: PE (Parity Error) • Bit 4: OWE (Overrun Error) • Bit 3: IDLE (Idle Line Detected) • Bit 2: TC (Transmission Complete) • Bit 1: RXNE (Read Data Register Not Empty) • Bit 0: TXE (Transmit Data Register Empty)

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Offset	Register	Access	Description
0x00	SR	Read-Only	Status Register (SR_out)
0x04	RDR	Read-Only	Read Data Reg (RDR_out). Reading triggers RDR_ren
0x08	TDR	Write-Only	Transmit Data Reg (TDR_in)
0x0C	CR1	Write-Only	Control Register 1 (CR_in)
0x10	BRR	Write-Only	Baud Rate Register (BRR_in)

Table 1: UART APB Wrapper Memory Map